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Prokaryome: A Comprehensive WDL-Based Workflow for Prokaryotic Genome Assembly, Polishing, Annotation, and Visualization

Saifeldeen M. Ibrahim [1,2¶]

1 Bioinformatics Department, The Agricultural Genetic Engineering Research Institute, ARC, Egypt **2** Biotechnology Department, Faculty of Agriculture, Al-Azhar University, Egypt ¶ Corresponding author

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Software

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Summary

`Prokaryome` is an open-source workflow designed to facilitate prokaryotic genome assembly, polishing, annotation, and visualization. Built using the `Workflow Description Language` (`WDL`), it streamlines the analysis of paired-end sequencing data from raw reads to polished, annotated genomes. The pipeline includes essential preprocessing steps such as quality control, sequence trimming, and error correction, as well as advanced assembly polishing and genome annotation. `Prokaryome` ensures high-quality genome assemblies by incorporating four rounds of polishing for the assembled reads to decrease the error rate of the assembly process, and provide visualizations and detailed reports of the assembly quality and the annotated features, to make it a comprehensive tool for microbiome research and comparative genomics.

Statement of need

Prokaryotic genome analysis is a very critical component of any microbiome research that can enable researchers to investigate the microbial diversity, functionality, and evolutionary relationships (Rajguru et al., 2024). However, this process is often fragmented and requiring multiple tools and workflows that demand a high level of technical expertise and high coding skills. So, many researchers face challenges in integrating these tools, ensuring compatibility, and maintaining reproducibility across their analyses. These challenges are further compounded by the need to process large datasets efficiently while maintaining high-quality outputs.

Traditional workflows for prokaryotic genome assembly and annotation typically involve manual execution of individual steps, such as quality control (Patel & Jain, 2012), trimming (Lee et al., 2021, pp. kong2011btrim, macmanes2014optimal), assembly (Wick et al., 2017), polishing, and annotation (Zimin & Salzberg, 2020). This not only increases the risk of errors but also makes the process time-consuming and difficult to replicate. Moreover, existing pipelines often fail to provide a seamless transition between steps or to include robust visualization features, which are critical for interpreting and presenting results effectively. Prokaryome, our proposed tool, addresses these challenges by integrating all essential steps into a single, streamlined pipeline. Designed with a focus on reproducibility, ease of use by all researcher, and comprehensive functionality. It designed by the Workflow Description Language (WDL) which is a domain-specific language designed to facilitate the development and execution of complex bioinformatics workflows efficiently (Van der Auwera & O'Connor, 2020). It takes the raw fastq sequencing data and automates the entire genome assembly process to the generation of annotated genomes and visualizations. As shown in Figure 1, the workflow begins with FastQC and MultiQC (Ewels et al., 2016), which assesses the quality of raw reads, followed by Trimmomatic, which trims low-quality bases (Bolger et al., 2014). The cleaned reads are then assembled into contigs and scaffolds using SPAdes (Bankevich et al., 2012), generating a

draft genome. The draft undergoes four iterative rounds of polishing with BWA, Samtools, and Pilon (Danecek et al., 2021; Jung & Han, 2022; Walker et al., 2014), enhancing assembly accuracy. QUAST evaluates the assembly quality against a reference genome (Gurevich et al., 2013), while RagTag scaffolds the polished assembly to produce a draft genome aligned to the reference (Alonge et al., 2022). The draft is further refined using a customized python script, annotated with Prokka to identify genomic features (Seemann, 2014), and lastly, the draft genome is visualized with Genovi, generating comprehensive reports and graphical summaries for downstream analysis (Cumsille et al., 2023).

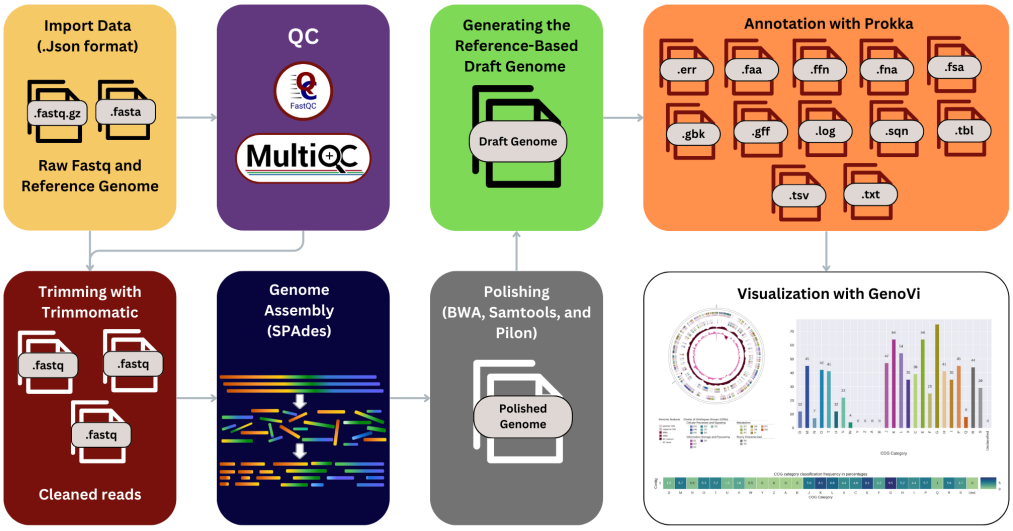

Figure 1: Prokaryome Workflow.

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